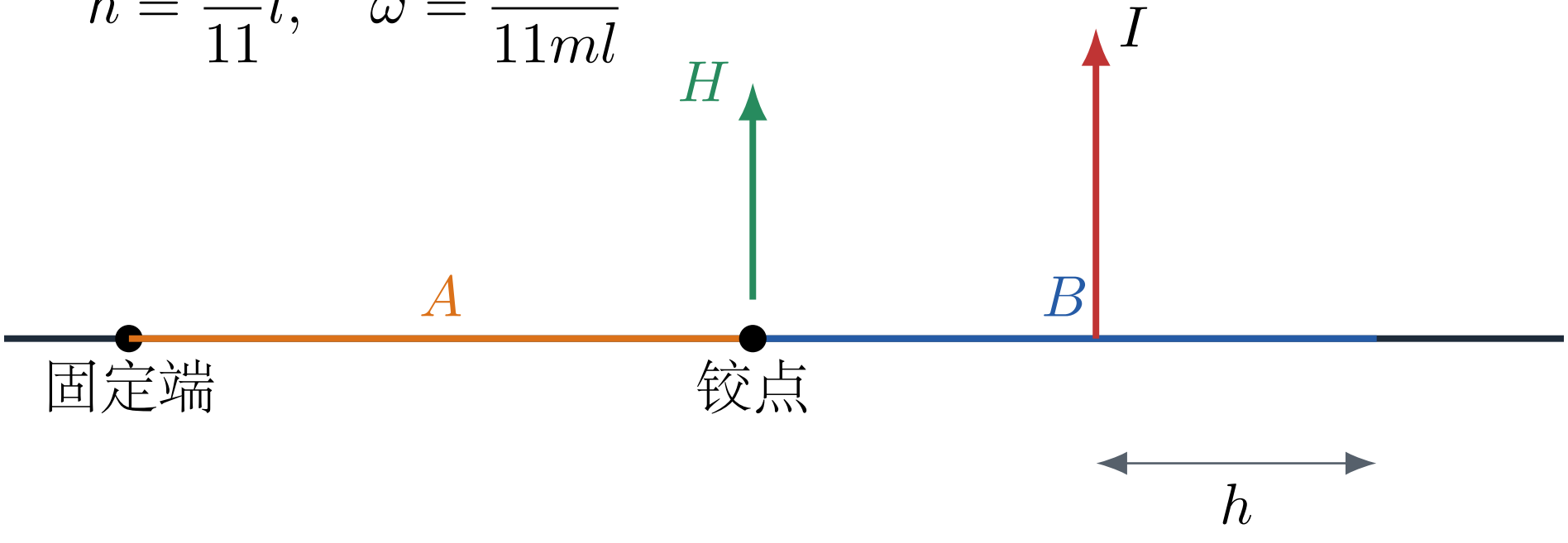
$$I + H = \frac{3}{2} ml\omega$$

$$-Hl = \frac{1}{3} ml^2\omega$$

$$h = \frac{6}{11} l, \quad \omega = \frac{6I}{11ml}$$